

MARATHWADA INSTITUTE OF TECHNOLOGY, AURANGABAD		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: Implementation of different types of SQL operators with suitable example 1) Arithmetic operators 2) Logical operators 3) Comparison Operators 4) Special operators 5) Set operators		
DEPARTMENT: COMPUTER SCIENCE AND ENGINEERING		LABORATORY: MIT-CSE-DBIT
EXPERIMENT NO.: 04		YEAR: 2017-18
Class: T.E.	PART: I	SUBJECT: Database Management System

AIM: Implementation of different types of SQL operators with suitable example

- 1) Arithmetic operators
- 2) Logical operators
- 3) Comparison Operators
- 4) Special operators
- 5) Set operators

THEORY:

SQL Operators	
Arithmetic Operators	+, -, /, *
Logical operators	AND, OR, NOT
Comparison Operators	BETWEEN, IN, LIKE, IS NULL, NOT LIKE, NOT IN, NOT NULL, NOT BETWEEN, =, >, >=, <, <=, <>,
Special Operators	DISTINCT, ALL
Set Operators	UNION, INTERSECT, UNION ALL, MINUS

You can use arithmetic operators in any clause of a SQL statement except in the FROM clause.

AND: Returns TRUE if *both* component conditions are true

OR: Returns TRUE if *either* component condition is true

SET OPERATOR: Set operators are used to join the results of two (or more) SELECT statements. The SET operators available in Oracle 11g are UNION, UNION ALL, INTERSECT, and MINUS.

UNION: The UNION set operator returns the combined results of the two SELECT statements. Essentially, it removes duplicates from the results i.e. only one row will be listed for each duplicated result.

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UNION ALL: All row selected by either query, including all duplicates

INTERSECT: All distinct rows selected by both queries

MINUS : Use the MINUS operator to return rows returned by the first query that are not present in the second query.

IN: Use the IN membership condition to test for values in a list.

LIKE: for pattern matching

% denotes zero or many characters.

_ denotes one character.

NULL: Test for nulls with the IS NULL operator.

(EXPECTED OBSERVATION / CALCULATION / RESULT)

Sample Output:

Query 1

Company chairman has decided to give increment of Rs. 1000 for all employees so update their salary accordingly

4 row(s) updated.

EMP_ID	FNAM	EMNAME	LNAME	DOB	ADDRESS	SALARY	GENDER
1001	Mahesh	S	Kumar	10-JAN-88	Pune	26000	M
1002	Sushma	Jagdish	Kale	02-MAY-87	Ahmednagar	16000	F
1002	Manisha	Rajiv	Joshi	25-FEB-78	Aurangabad	36000	F
1003	Sandeep	Kamal	Mishra	04-APR-90	Aurangabad	41000	M

Query 2

Display employee id, employee first name, salary whose salary is greater than 10000 and less than 30000.

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EMP_IDFNAMESALARY

1001 Mahesh 26000
1002 Sushma 16000

Query 3

Display all addresses where starting with A

ADDRESS
Ahmednagar
Aurangabad

Query 4

Display details in terms of department ids for which an employee has worked for or working for
Deptment no 101,104,102,103

EMP_ID	DEPT_NUM
1001	101
1001	104
1002	101
1002	102
1003	101
1004	102
1004	103
1005	104
1006	103
1007	103

Query 5

Display last name of all employees who are born in year '1978'

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LNAME

Joshi

Query 6

Find employee details whose commission amount is 20% of its salary

EMP_ID	FNAME	SALARY	COMMISSION
1002	Sushma	16000	3200

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